Senior Design Project
END-TERM EVALUATION

Krishi Sahayata: Simplifying Insurance for farmers with seamless local language support

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Presentation Outline

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Introduction

Project Overview

- Krishi Sahayata is a web-based platform designed to simplify crop insurance settlements for Indian farmers by reducing delays and improving transparency.
- The traditional process involves manual verification and paperwork, causing lengthy waiting periods before farmers receive compensation.
- Fraudulent claims and lack of real-time data verification result in stricter policies, further delaying genuine payouts.
- Many farmers face difficulties due to language barriers and low awareness, making it challenging to understand policies and file claims.
- By integrating an AI-powered chatbot, multilingual support, and real-time weather-based verification, Krishi Sahayata ensures a faster, transparent, and more accessible claim settlement process.

Introduction

Problem Statement

- Indian farmers face language barriers, low awareness, and a slow, complex insurance process, making claim settlements difficult. Lack of transparency breeds mistrust, while fraud leads to stricter rules, delaying genuine payouts.
- Financial distress from crop failures often pushes farmers into debt. Krishi Sahayata addresses these issues with local-language support, transparency, and fraud prevention, ensuring a smooth and fair claim process.
- This solution helps farmers secure their livelihoods, strengthens economic resilience, and promotes sustainability.

Introduction

❑ Objectives & Expected Impact

- **Simplify Crop Insurance** – Streamline the claim process for farmers, government agencies, and insurers.
- **AI Chatbot Assistance** – Guide farmers in understanding policies, filing claims, and tracking status.
- **Multilingual & Voice Support** – Provide accessibility in 12 Indian languages with voice-over.
- **Fraud Prevention** – Use meteorological data to verify claims and eliminate false claims.
- **Real-time Updates** – Notify farmers about claim status via SMS and app tracking.
- **Financial Security** – Ensure timely, fair settlements to reduce distress and debt.
- **Educational Support** – Offer video tutorials and farming advice through Krishi Mitra chatbot.
- **User-Friendly Interface** – Enable easy navigation for farmers with minimal digital literacy.

Background & Related Work

❑ Existing Solutions

- **Pradhan Mantri Fasal Bima Yojana (PMFBY)**: A government-backed insurance scheme offering financial aid for crop losses due to natural calamities. However, it faces challenges like slow claim settlements, lack of awareness, and a tedious verification process.

- **Kshema General Insurance Limited**: A digital platform for purchasing crop insurance with faster claim settlements. While it simplifies the process, it may still not fully address fraud, language barriers, and accessibility for rural farmers.

Background & Related Work

Improvements over existing solutions

- Eliminating Traditional Paper-Based Verification
- Faster and Transparent end to end Settlements
- AI-Powered Multilingual Chatbot
- Fraud Prevention to ensure claim verification with Data-Driven Insights
- Direct Support Without Middlemen

Proposed Solution & Architecture

❑ Workflow Diagram

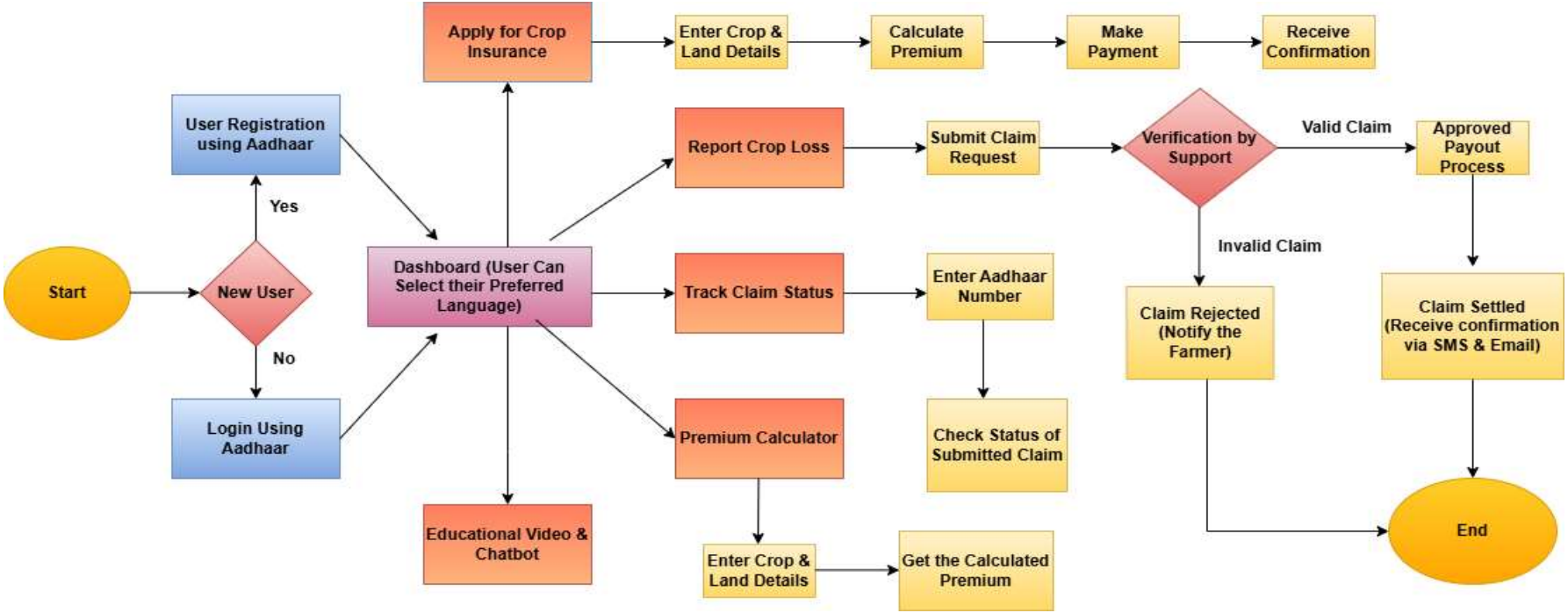

Figure 1 Flow diagram of Proposed solution

Proposed Solution & Architecture

❑ ER Diagram

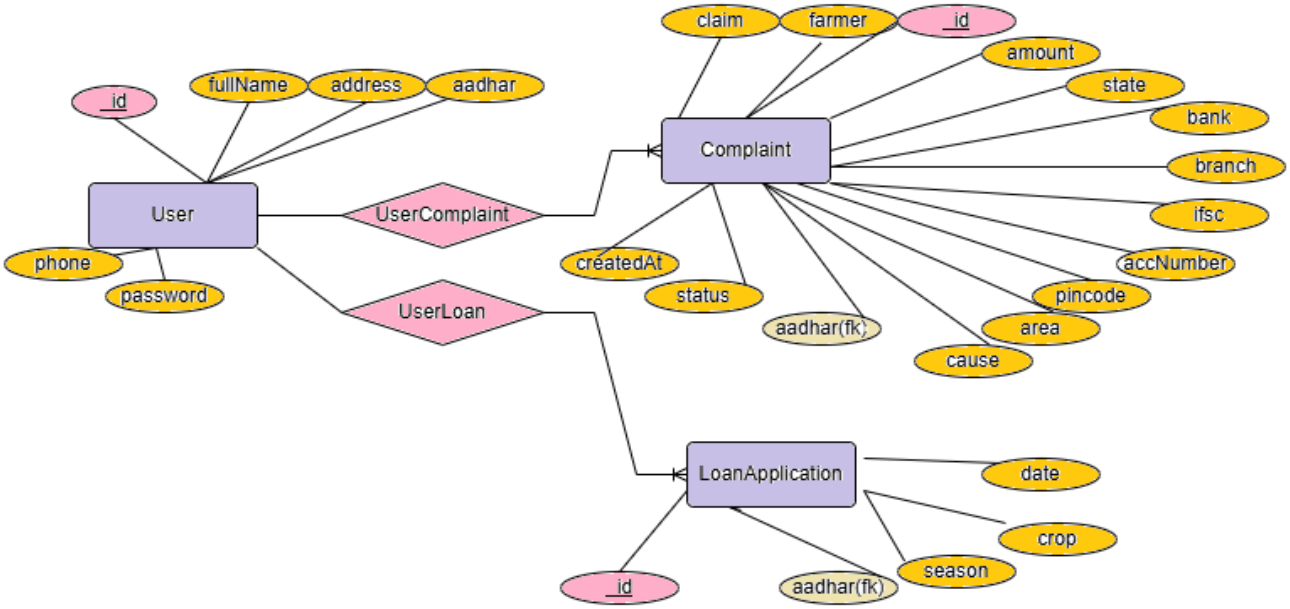

Figure 11 ER diagram of Proposed solution

Proposed Solution & Architecture

Features

- **Multilingual Support & Keyboard** – 12 Indian languages with voice and typing assistance
- **Contact Support** – Queries via app, response through email/phone
- **Premium Calculator** – Estimates insurance cost before applying
- **Report Crop Loss** – Quick claim filing for natural calamities
- **Admin Verification** – Officials verify claims using weather data
- **Real-Time Updates** – SMS notifications on claim status
- **Aadhaar-Based Tracking** – Farmers track claims using Aadhaar
- **Educational Resources** – Farming related Videos for guidance
- **Support & Guidance** – AI Powered Chatbot (Krishi Mitra)

Implementation Details

Technologies, Frameworks & Tools Used

- **Frontend** : React Js, Tailwind CSS
- **Libraries** : React-UI, Lucide-React, Ant-d
- **Backend** : Node Js, Express Js, Python
- **Database** : MongoDb
- **Tools** :VSCode, GitHub, Gemini 1.5 Pro, Postman
- **APIs** :Twilio, OpenMeteo, Translation, Text-to-Speech, Indic Keyboard, Razorpay

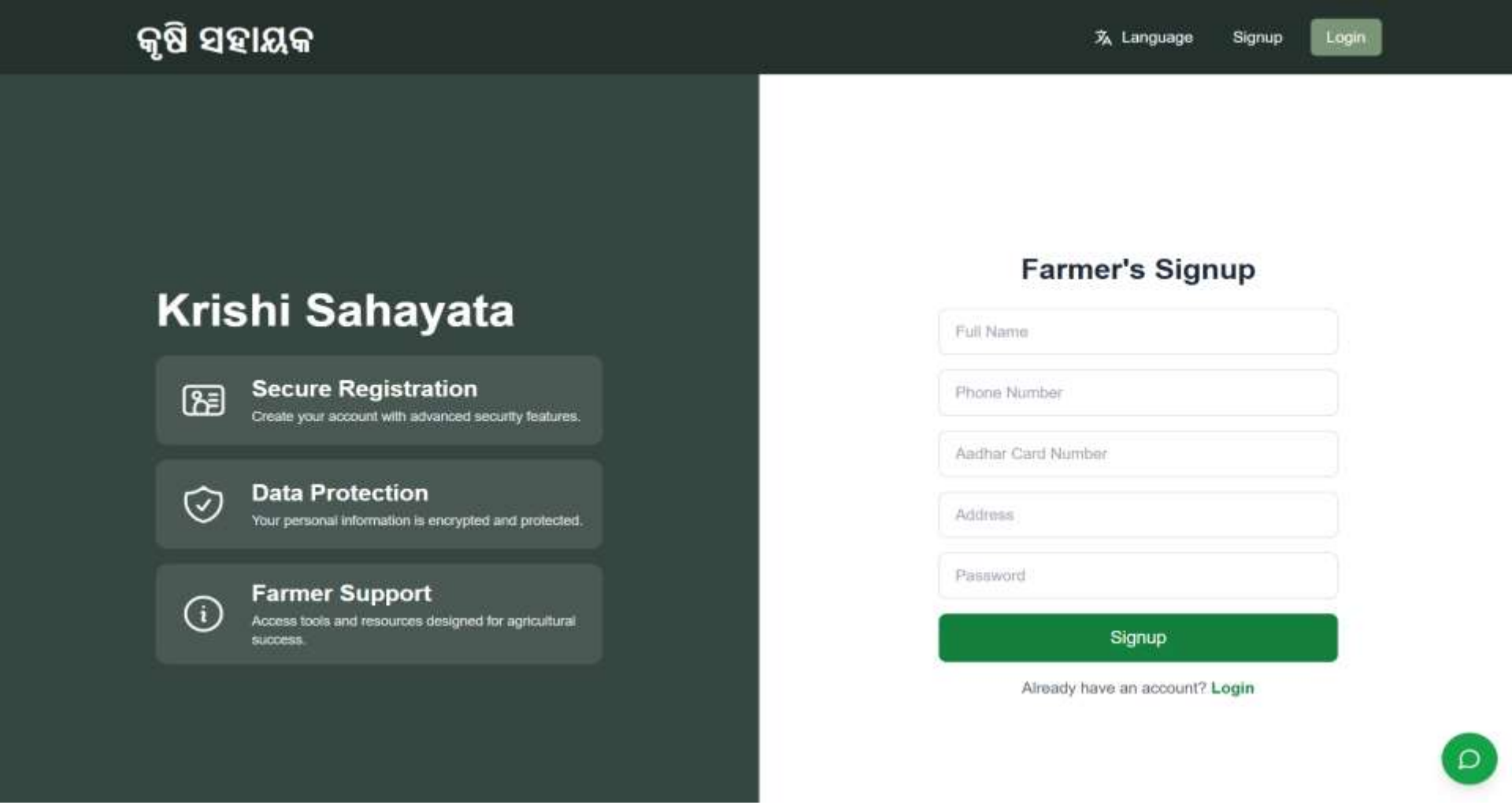

Figure 3(a) Login Page

Results and Analysis

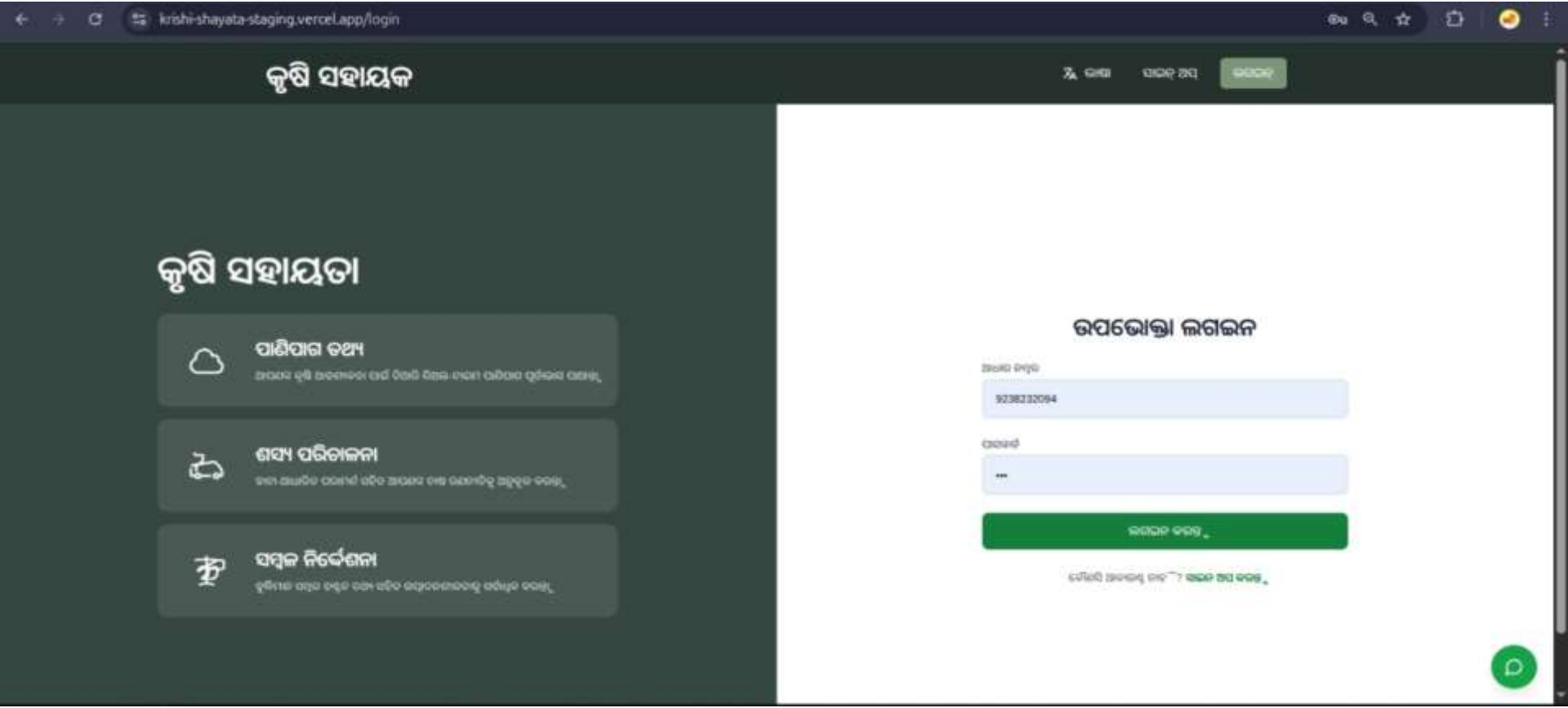

Figure 3(b) Login Page

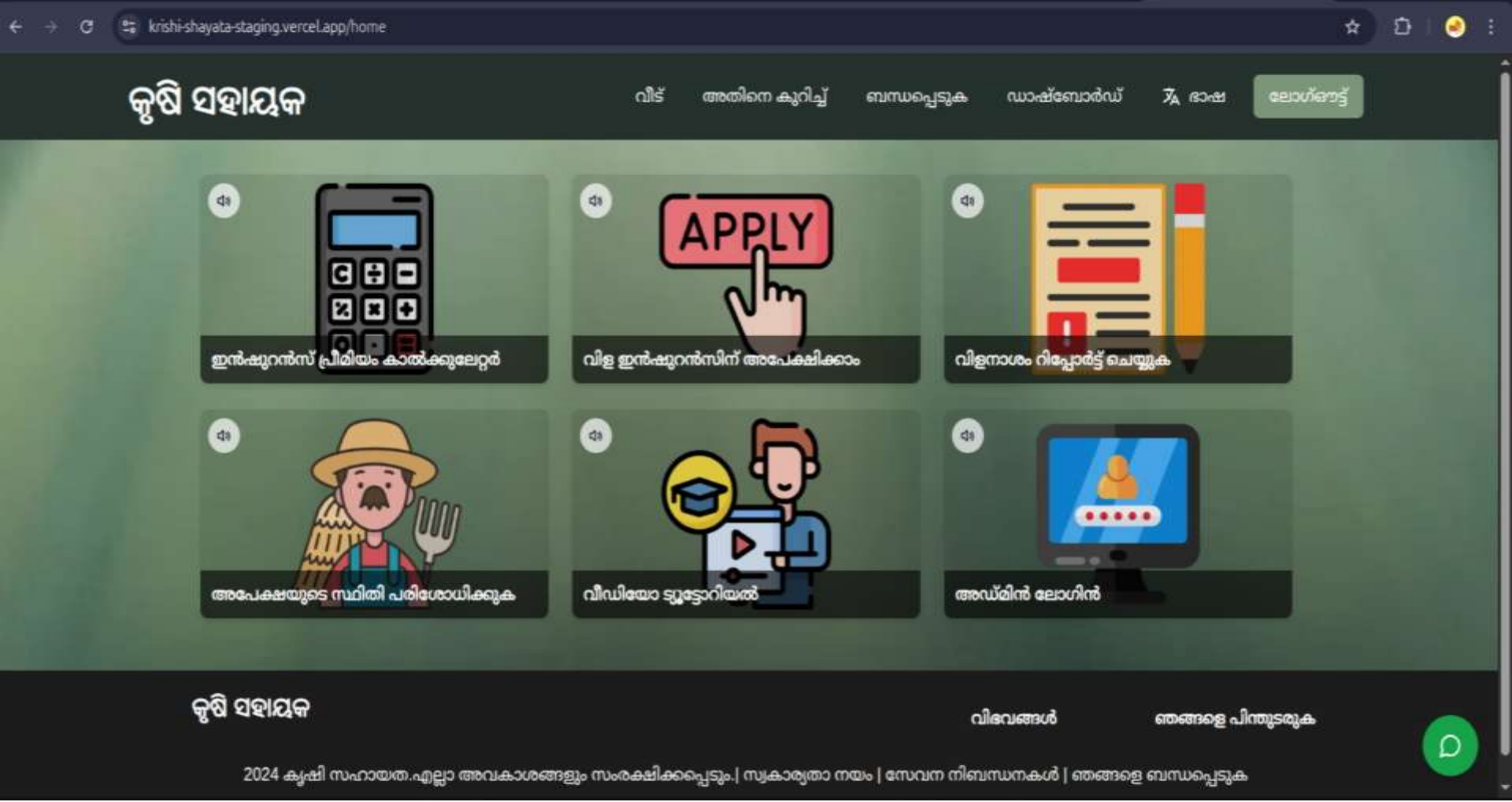

Figure 4 : Home Page

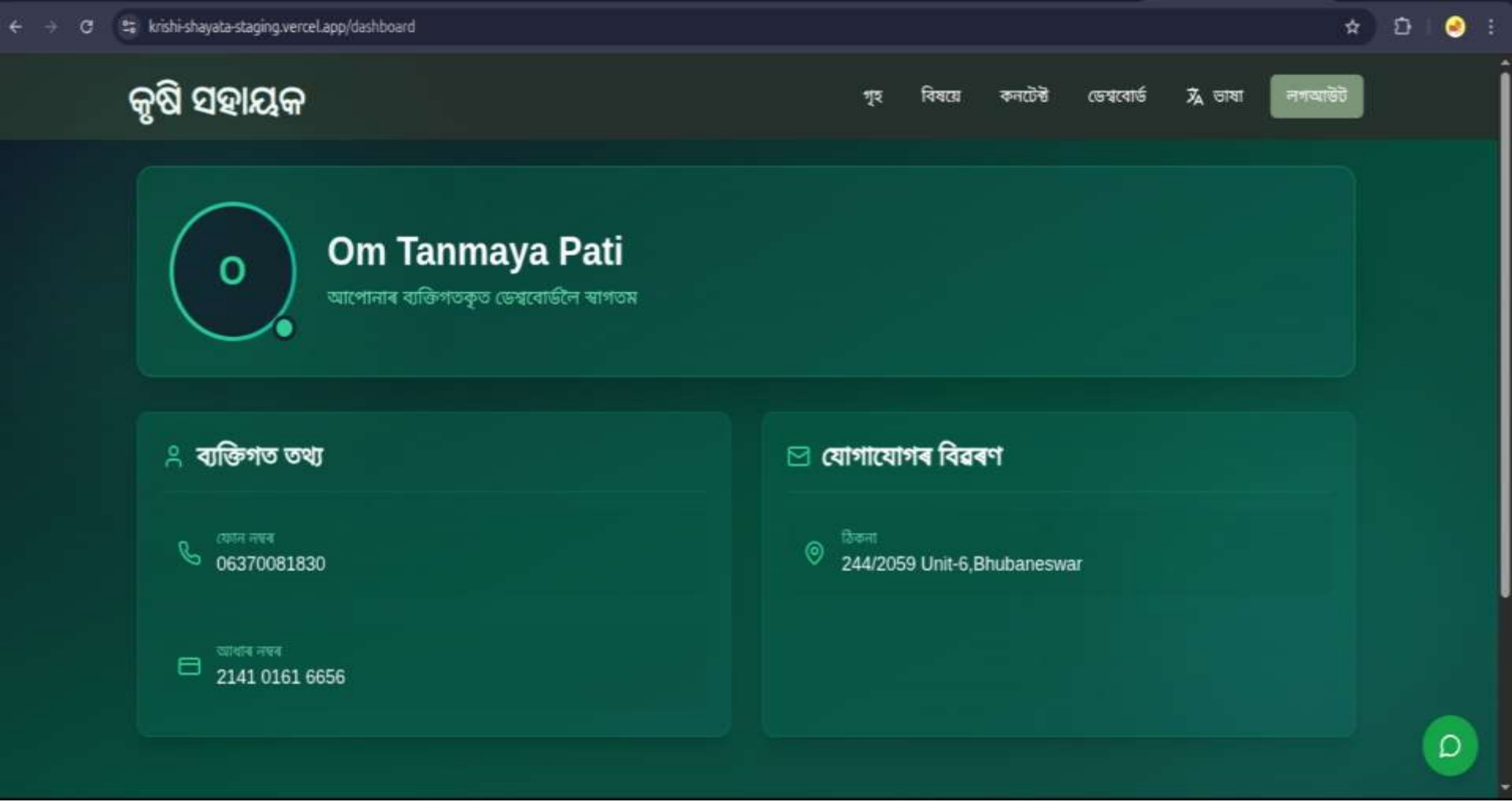

Figure 5 : Dashboard Page

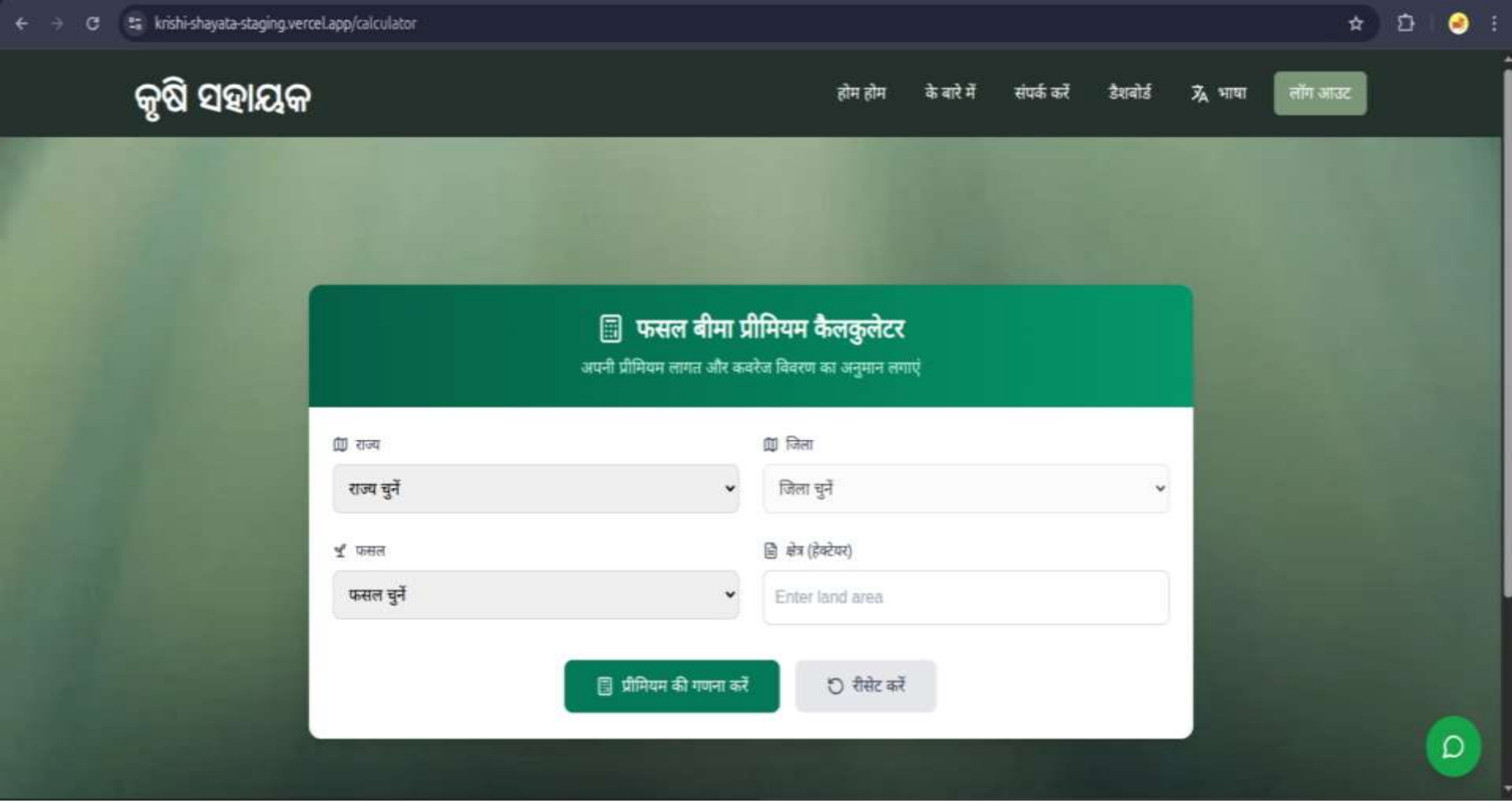

Figure 6 : Insurance Premium Calculator

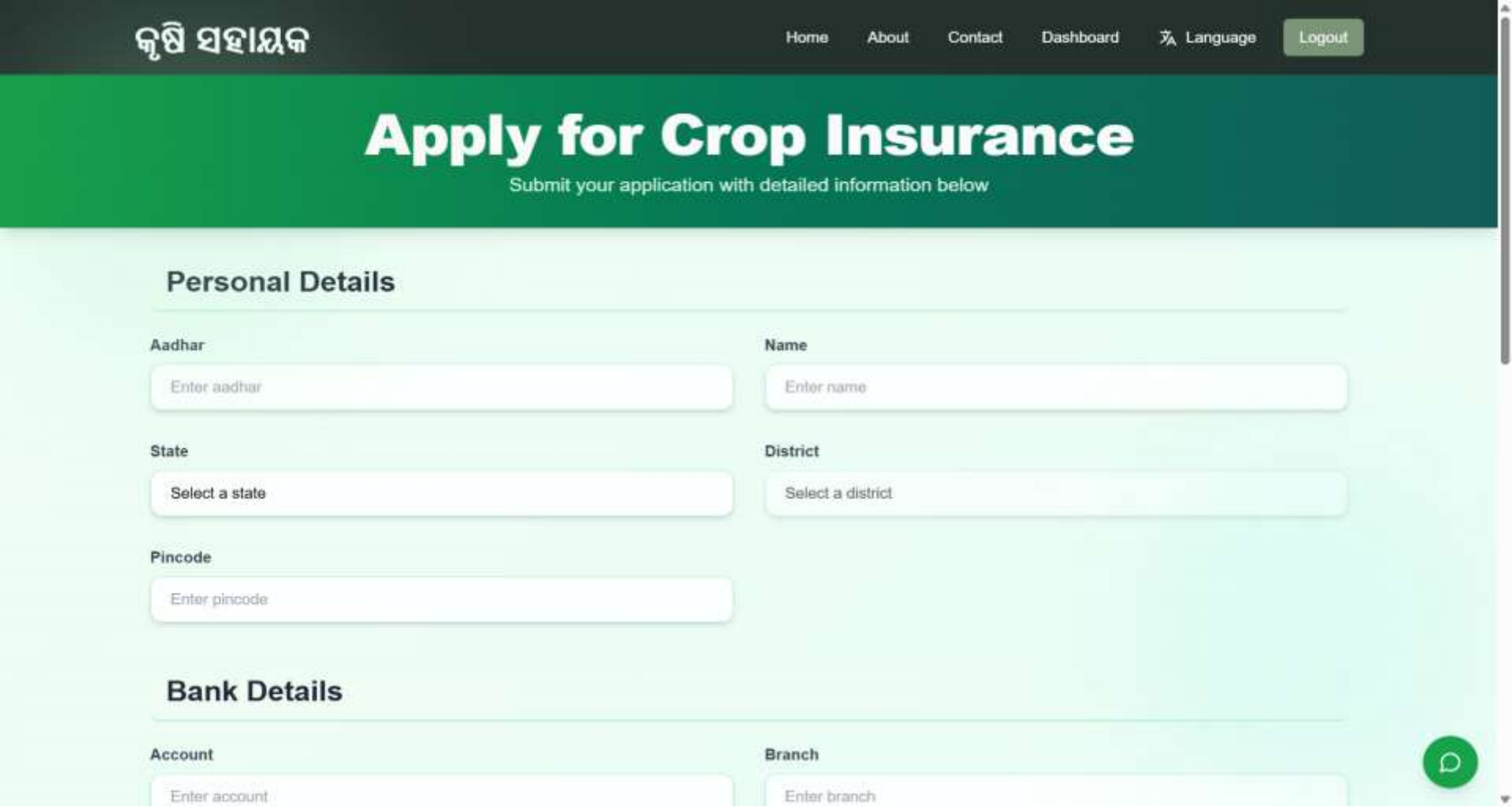

Figure 7 : Crop Insurance Application Form

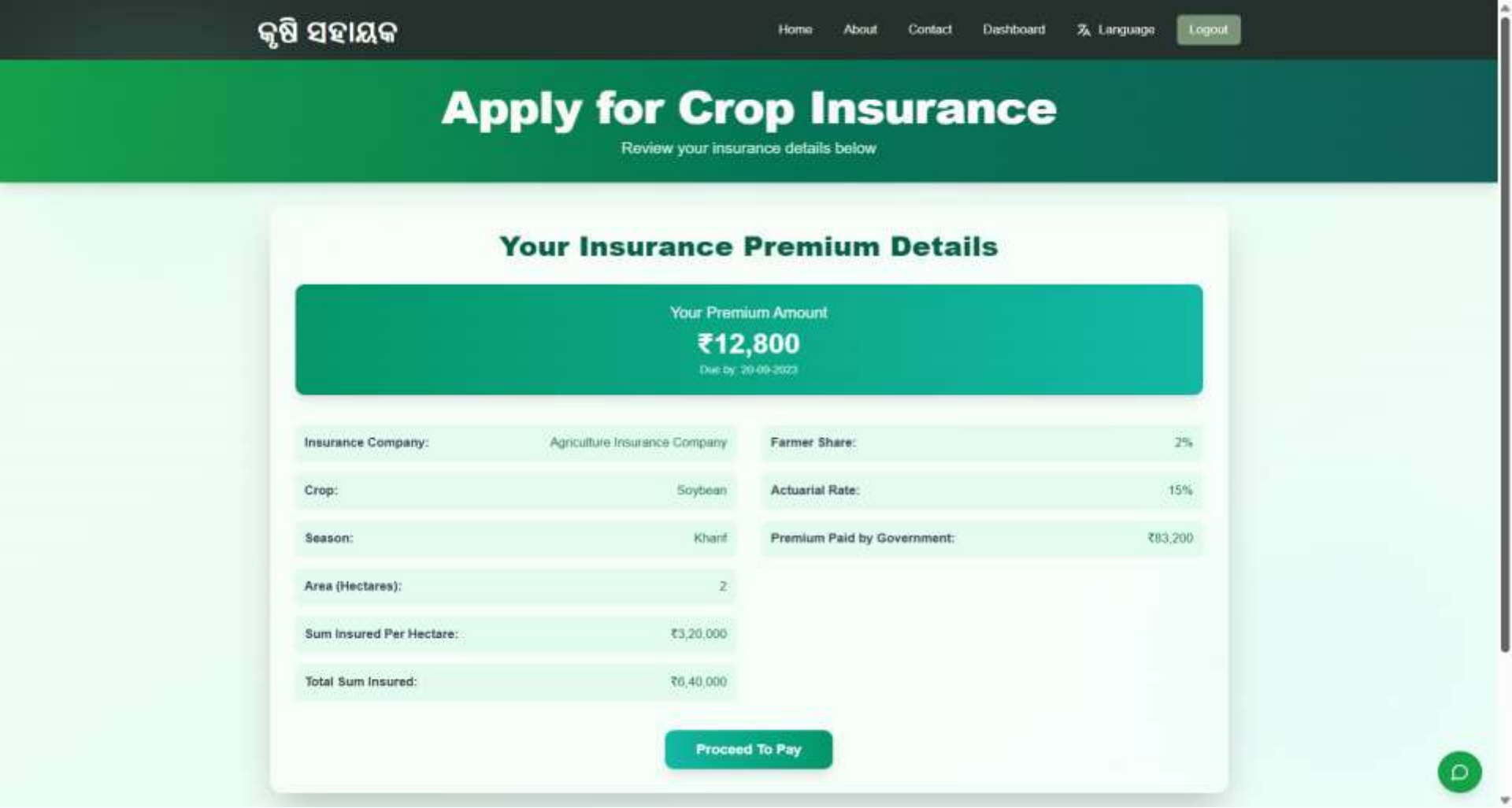

Figure 8 : Payment Details Page before Checkout

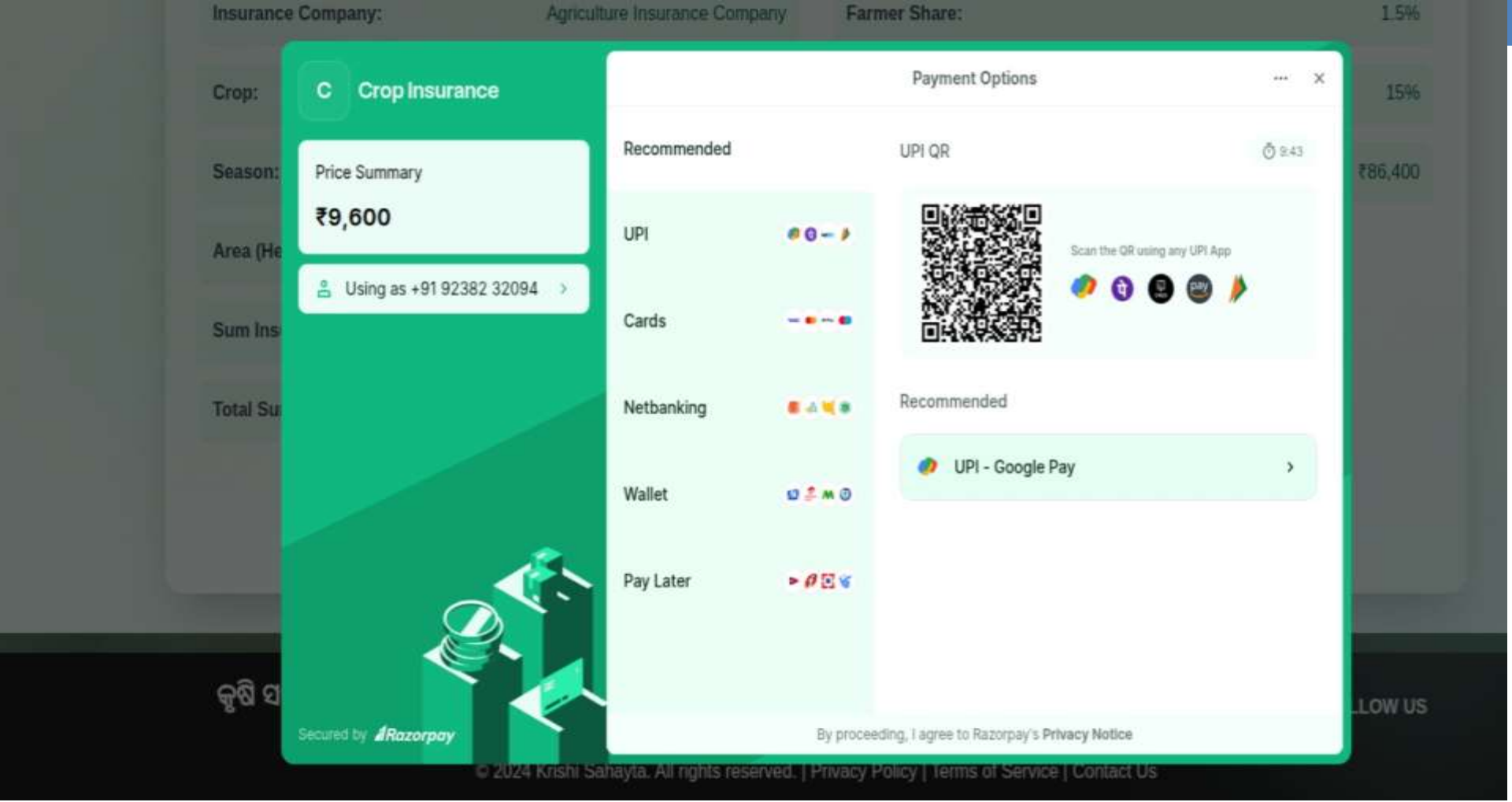

Figure 9 : Payment Checkout Page using Razorpay Gateway

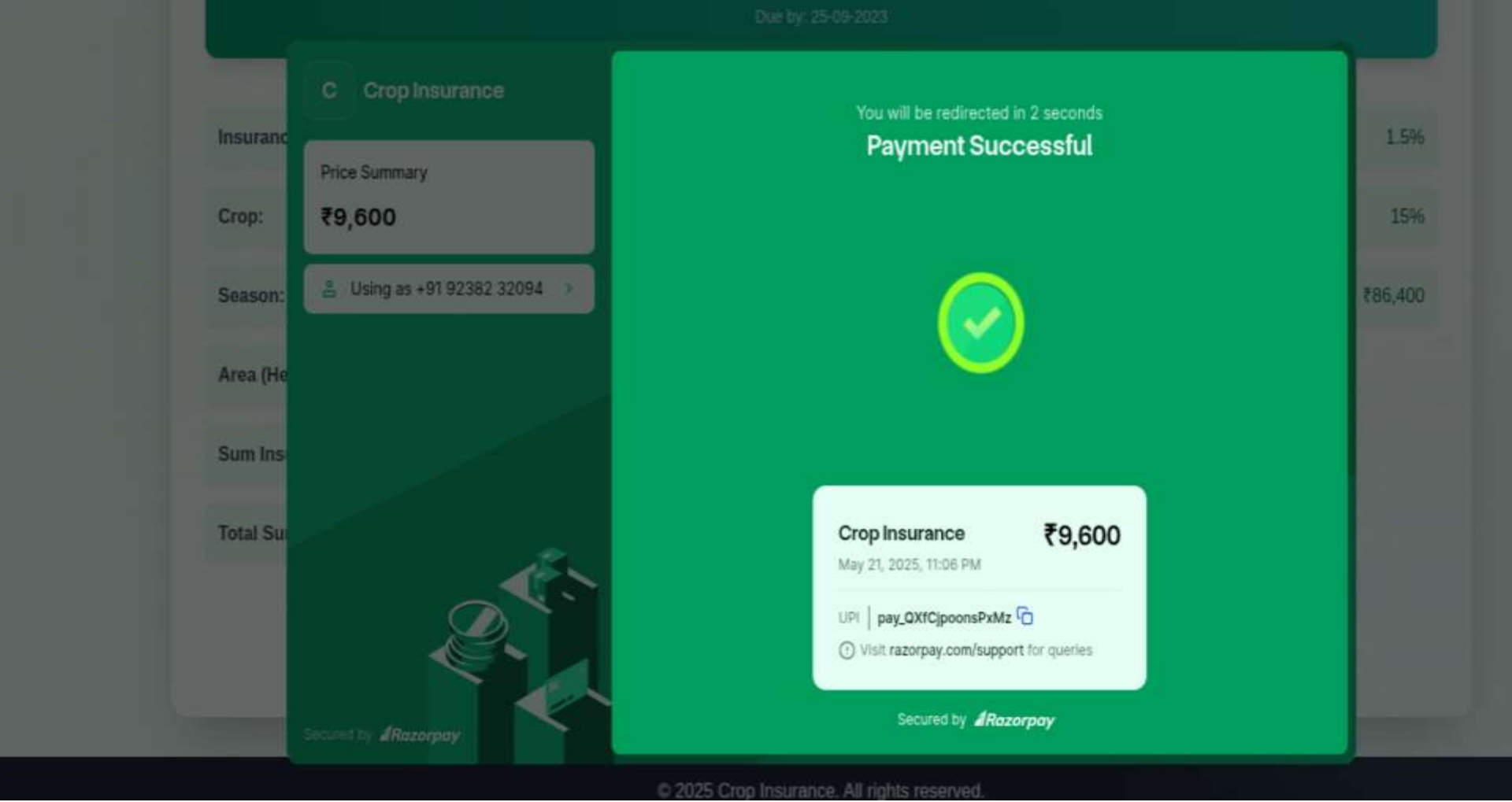

Figure 10 : Payment Successful Notification

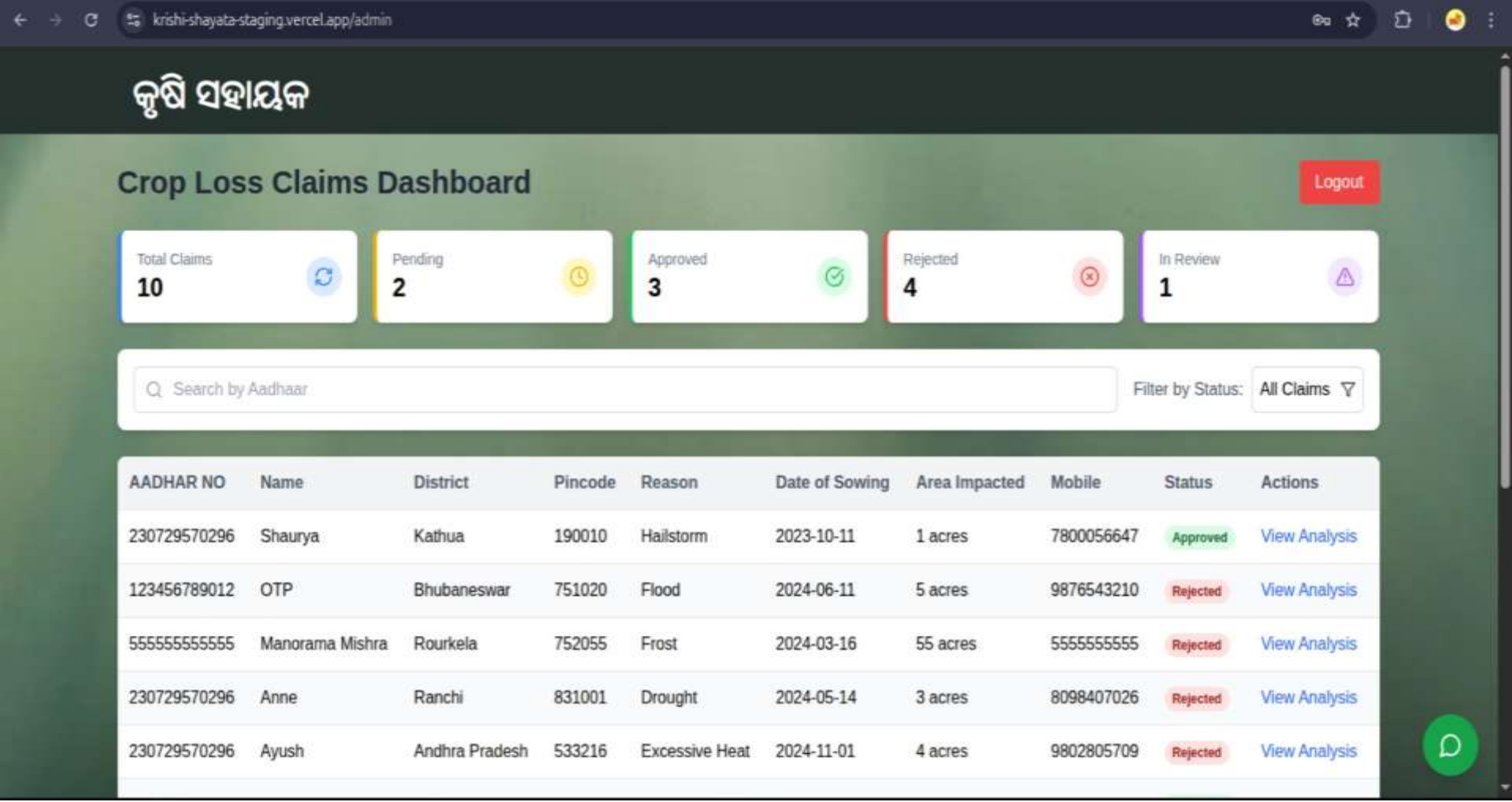

Figure 11 :Admin Dashboard where admins can review the claims

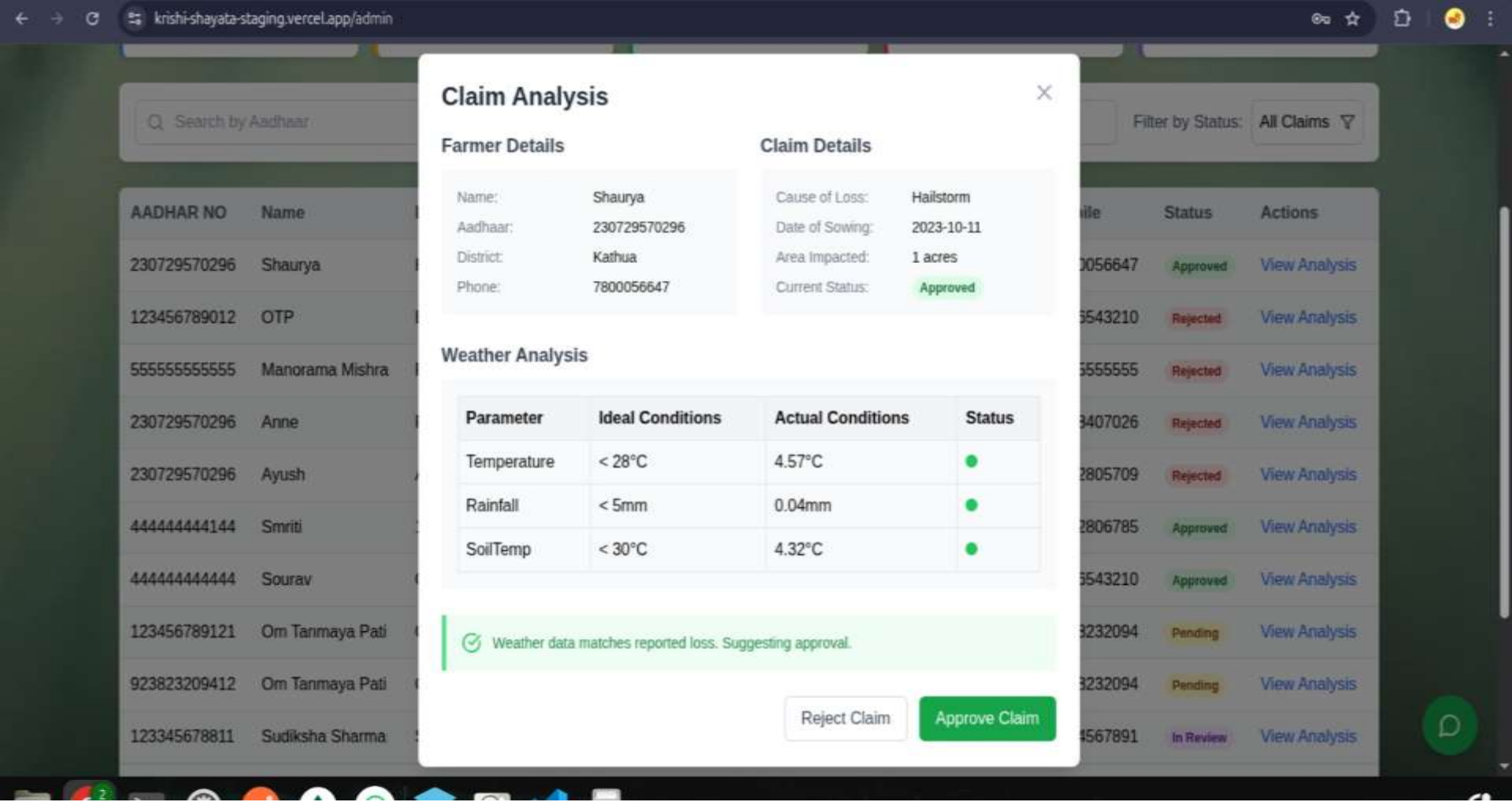

Figure 12 : Claim being verified by the weather APIs

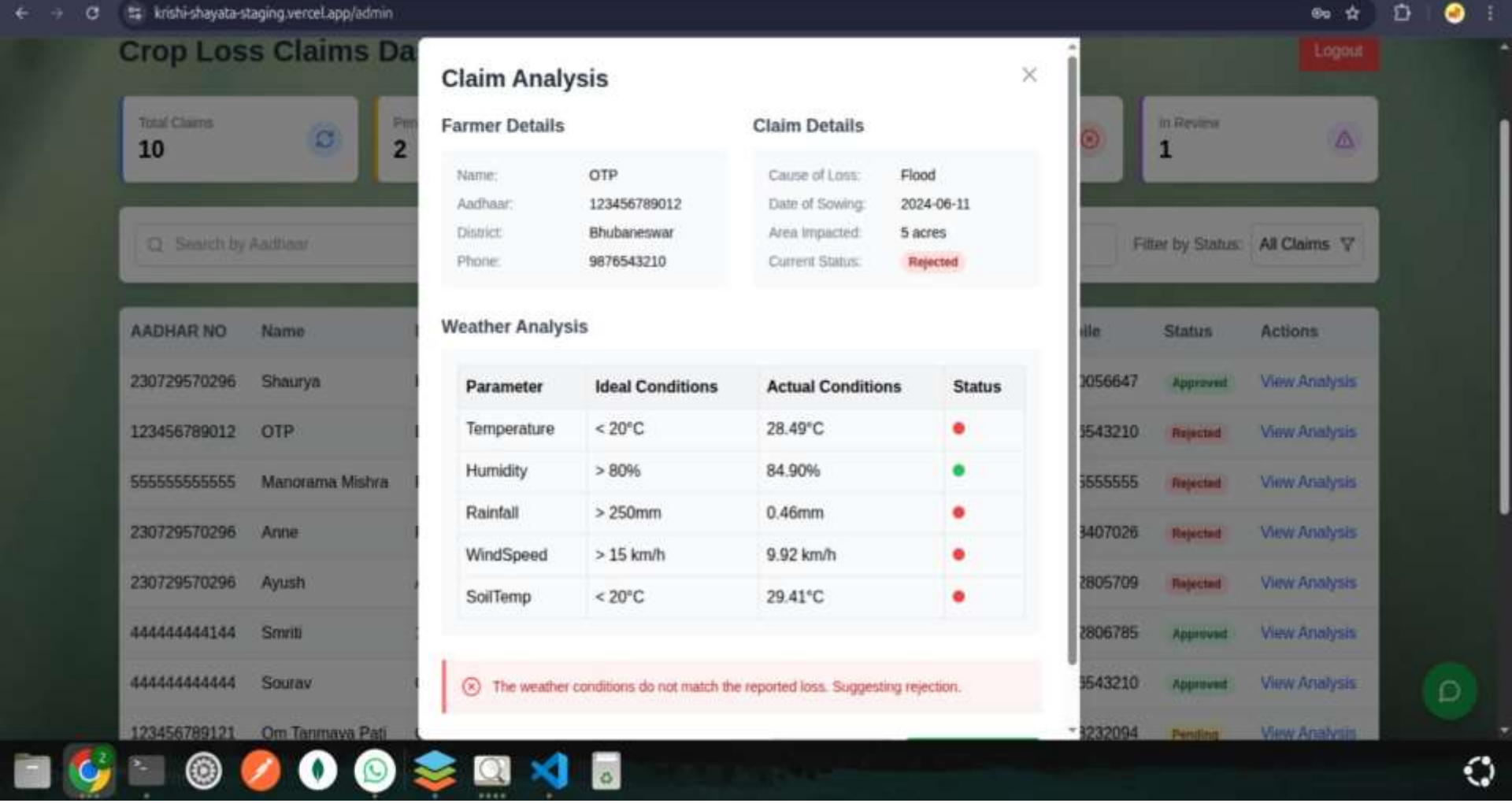

Figure 13 : Claim Rejected

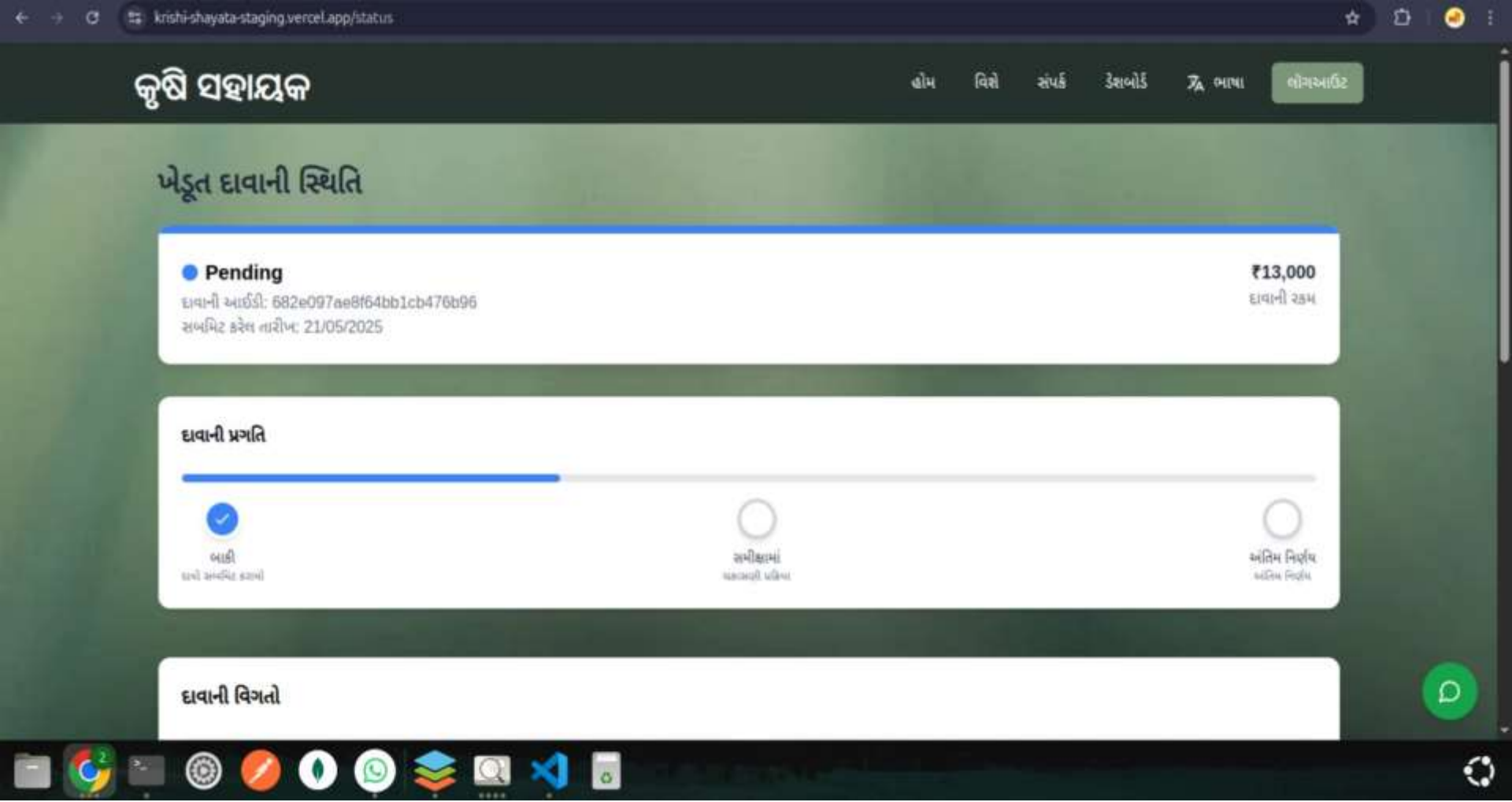

Figure 14 : Check Application Status for Farmers

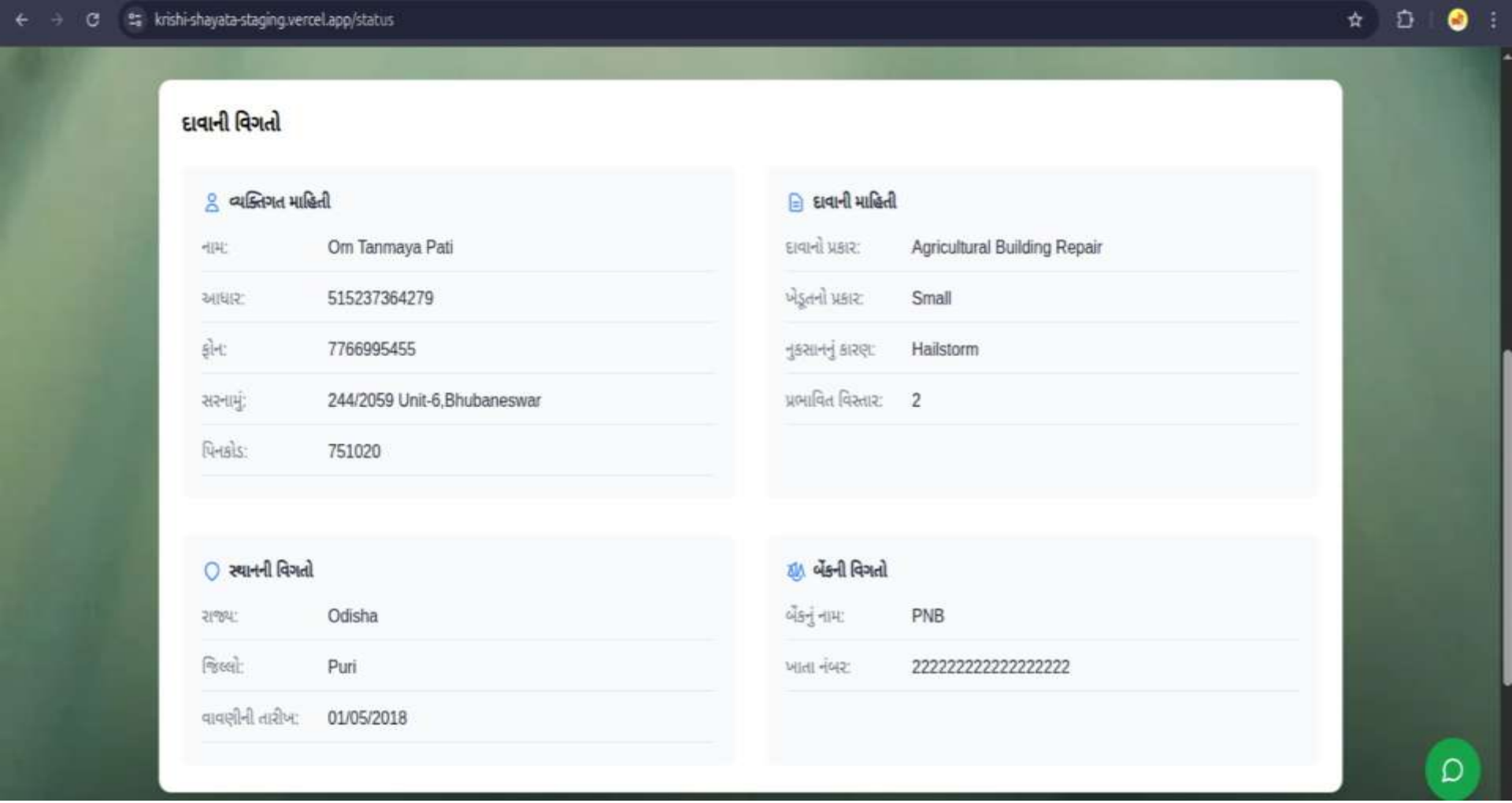

Figure 15 : Claim Application Details

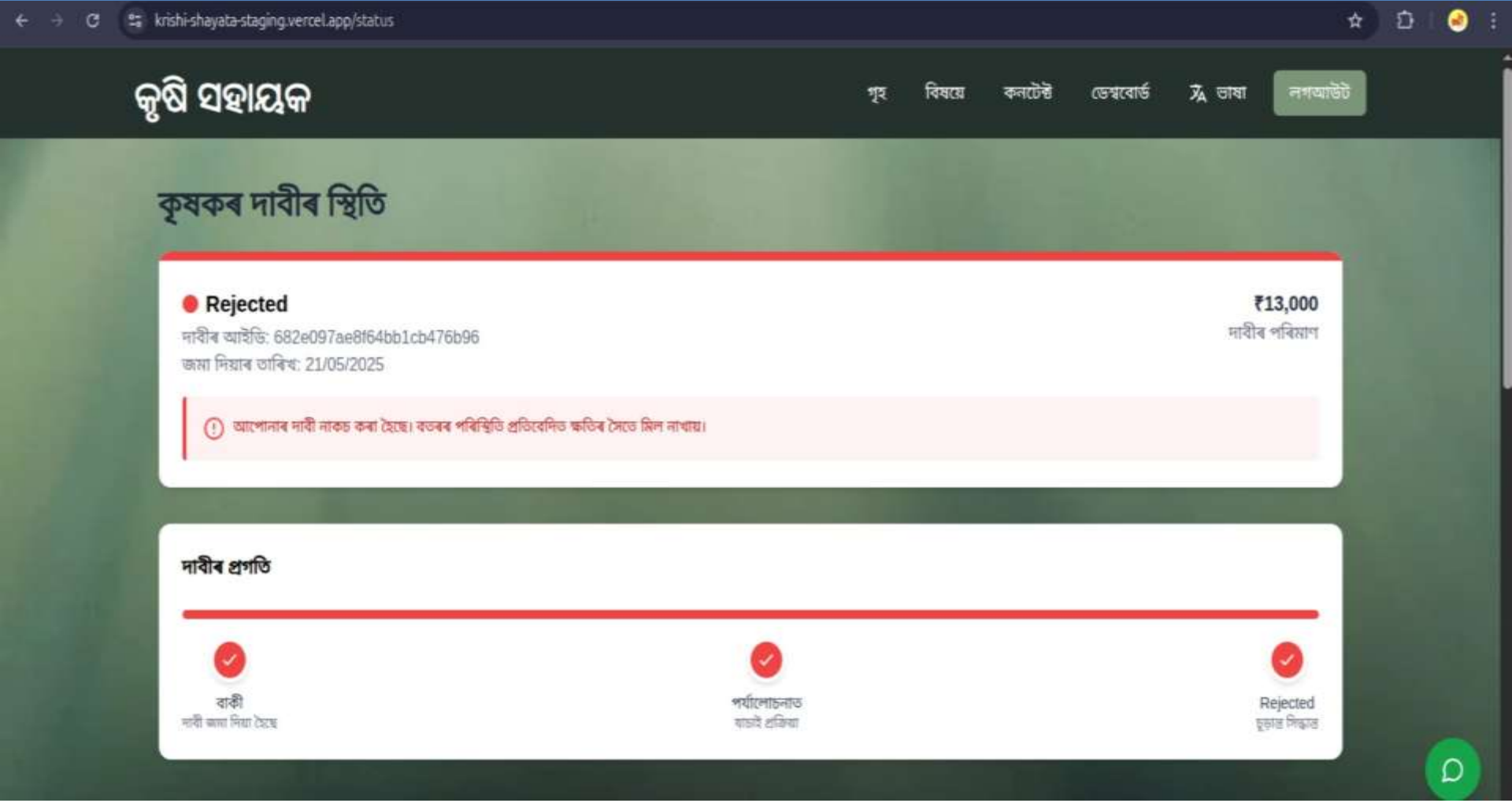

Figure 16 : Final Status of the Claim

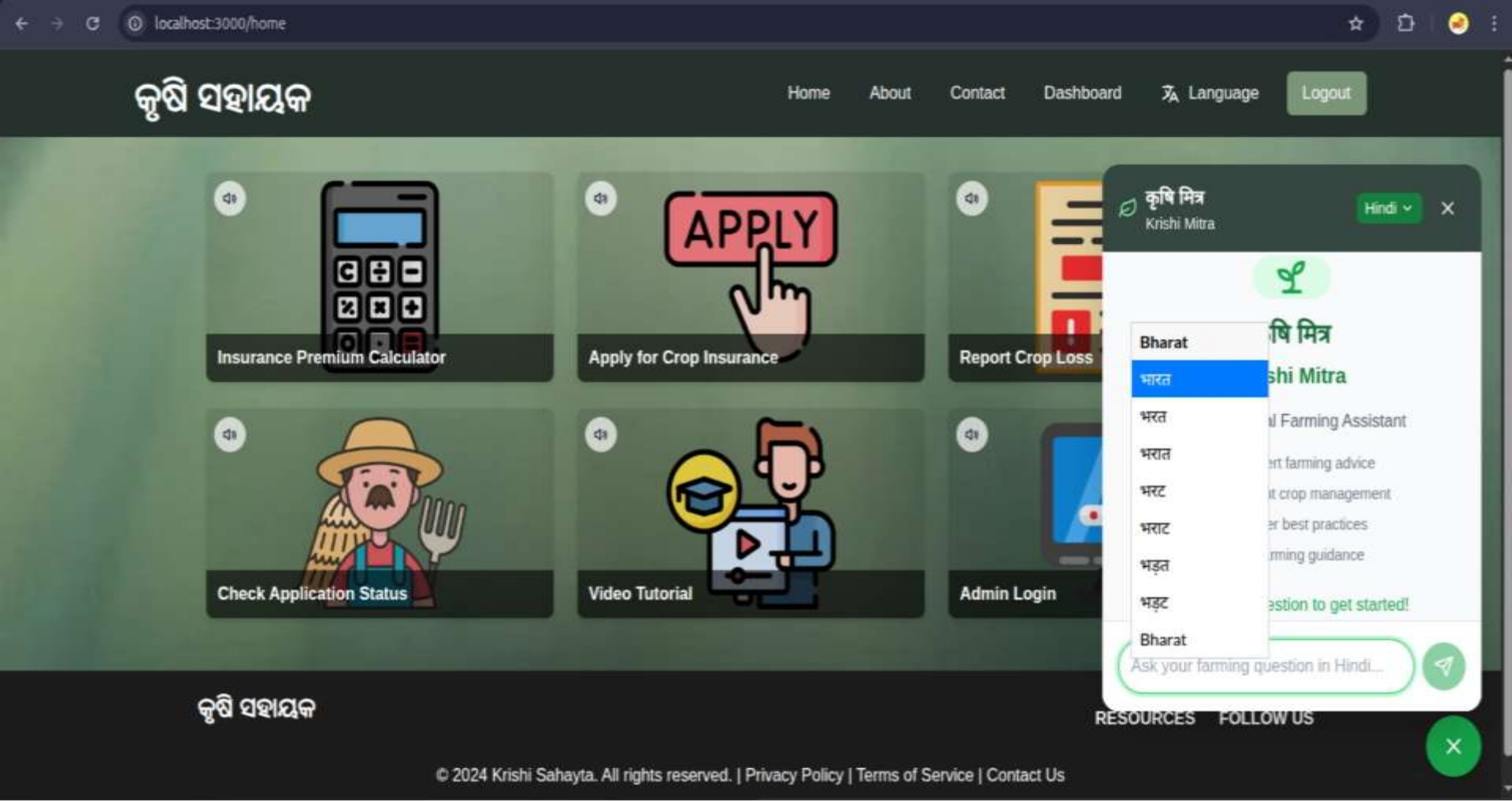

Figure 17 : Indic Keyboard inside the Krishi Mitra : ChatBot

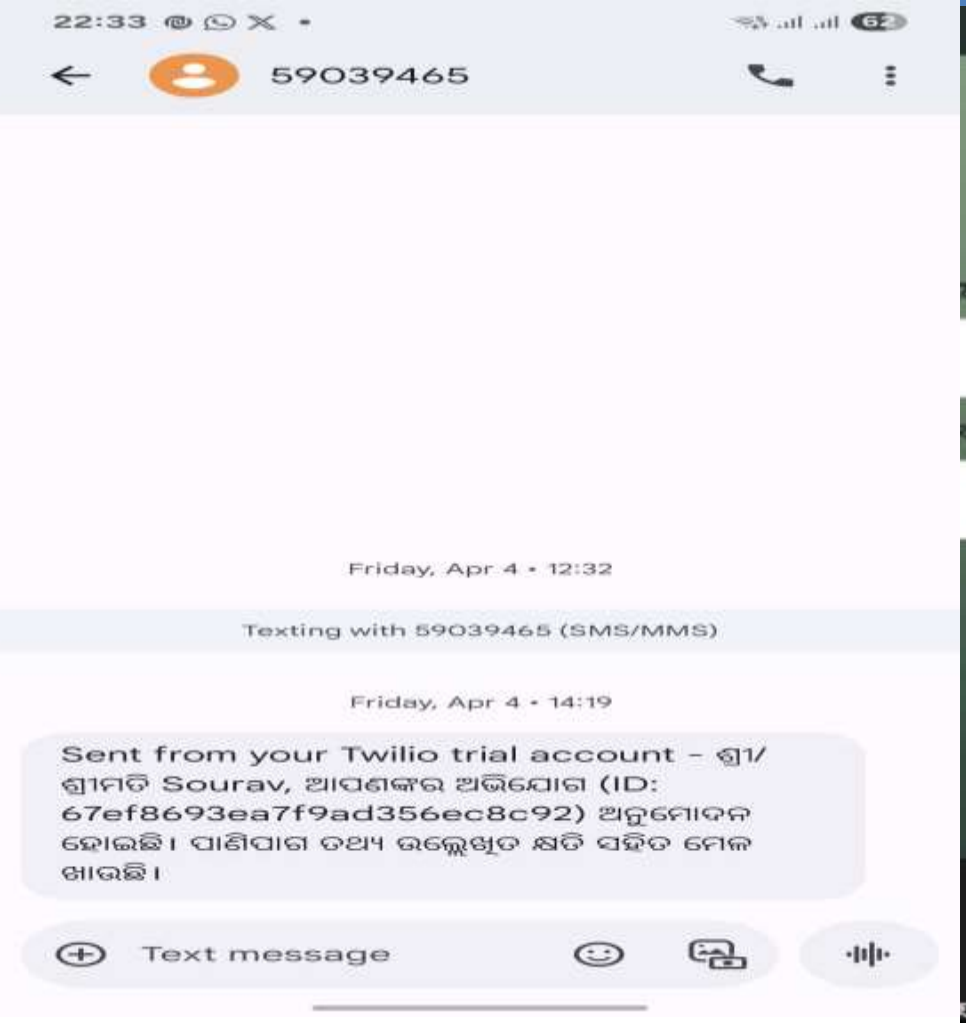

Figure 18(a): SMS Updates being received on Mobile Number

Figure 18(b): Krishi Mitra Assisting with Farming related queries

Bibliography

- Pradhan Mantri Krishi Fasal Bima Yojna : https://pmfby.gov.in
- Kshema General Insurance Limited : https://kshema.co/
- Open Meteo Weather API : https://open-meteo.com/
- Open Street Map API : https://www.openstreetmap.org/
- Gemini API : https://aistudio.google.com/apikey
- React Documentation: https://react.dev/
- Tailwind CSS Documentation : https://tailwindcss.com/

Thank
you